

Adaptive Project Planning for Epics, Features and User Stories

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Abstract—As per research reports, 28% of projects that follows a conventional waterfall project management model fail to deliver the requirements. This raises questions about the relevancy of conventional project management strategy in today's fast-paced business environment. Waterfall project management fails to meet end goals due to ever-changing market requirements and continuous advancements in technology. Thus there is a need to have a project management strategy that can adapt to a continuously changing environment. In this paper, we will be reviewing agile project management and its merits over conventional project management strategy.

Keywords: adaptive project management, work breakdown structure, themes, epics, features, user stories

I. INTRODUCTION

Most of us are aware of the fact that traditional project management has a static and straightforward strategy. For example, in a waterfall process, the project goes through five phases - requirement gathering, designing, implementation, testing, and deployment. Based on the complexity of the project, we possibly may need months or years to complete the project. In the fast-paced business environment and increasing demands of the market, our final product from the project may become obsolete at the time of completion. This may not only happen due to changes in requirements but also advancements in technology. So, to overcome these changing demands and to consistently respond to changes with the development of technologies, there is a need for an adaptive project framework.

Agile is all about innovation and nowadays most companies operate in highly dynamic environments, which mean they need tools that can adapt quickly to changes. They do not need just new products and services but also innovation in their business processes.[2]

II. ADAPTIVE PROJECT FRAMEWORK

A project's unknown aspects can arise at any time, and the adaptive project framework (APF), also known as adaptive project management (APM), takes this into account. Teams are more equipped to respond to the unexpected and anticipate it. Consider its central tenet to be "learning by doing."

Project scoping, cycle planning, cycled build, client check-point, and final review are the five stages of the APF, each having its own set of sub-stages and distinct deliverables. Although there are strict guidelines you must adhere to during these stages, the framework as a whole lives up to its reputation as a versatile tool. Due to its

adaptability, the project team can learn from its mistakes, solve problems, and involve the client wherever possible.

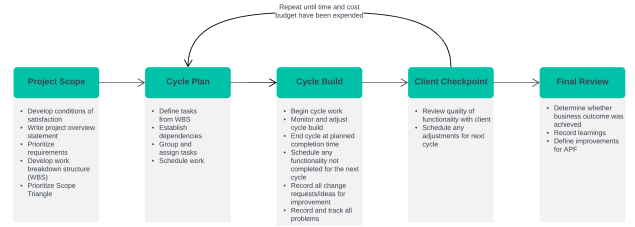


Fig. 1. Life Cycle in Adaptive Project Framework

III. WORK BREAKDOWN STRUCTURE

A project is visually, hierarchically, and deliverable-focused disassembled using a work breakdown structure (WBS). It is a useful diagram for project managers since it enables them to deconstruct the scope of their projects and see all the tasks necessary to finish them.

The work breakdown structure chart is a crucial project planning tool because it lists every phase of a project's work. The WBS levels below the project scope represent the activities, deliverables, and work packages required to complete the project from beginning to end. The final project deliverable, along with the tasks and work packages connected with it, lie on top of the WBS diagram.

A. Themes/initiatives

Initiatives offer another level of organization on top of epics. Generally, initiatives compile epics from several divisions of the company to achieve much bigger and broader goals than the epics themselves. Initiatives are often completed in multiple quarters or a year.

B. Epics

An Epics are large bodies of work delivered over several sprints. These initiatives bring strategic value and frequently have business cases behind them. Organizations can structure their work and continue to make progress toward important goals while breaking it down through the use of epics.

Agile epics resemble epics in literature or film in some ways. Each epic can be divided into a number of features, which are individual pieces of work. They are divided or sized in accordance with client or end-user needs as part of the Agile process. [5] In order to complete complex projects and consistently provide value, it is important to divide work into manageable deliverables.

For Example, Introducing a brand-new mobile app in conjunction with an online cosmetics store is the epic. A team of app developers will be put together to work on the numerous user stories, some of which might be-

- Features for augmented reality that allow users to virtually experiment on makeup.
- Chatbot features to help with simple questions.
- Special offers and discounts for devoted clients.

The mobile app can be tested and made ready for launch once all the user stories have been finished.

Benefits of epics

- **Better organization-** You may gather all of your user tales in one place and keep track of your ideas with the aid of epics. This facilitates project management and helps you stay on top of important requirements.
- **Improved time management-** Making an efficient project timetable is made simpler by splitting epics into sprints. Your time estimate will be even more accurate if tale points are assigned to indicate the sprint difficulty.
- **Clear client priority-** When agile teams are completely aware of the needs of the client, they produce goods of greater quality. There shouldn't be any doubt about the final deliverables because there are several user stories detailing explicit criteria.

Writing an Epic - In case of any doubts during the development work, a well-written epic will help provide a solid basis for understanding. Team members and stakeholders benefit from avoiding conflicts and misunderstandings. When drafting your Epic, it's extremely important to collaborate since this will be what you refer to when breaking down the work. An Epic encapsulates the future state of the user, the product, and the design requirements in the form of a story.

C. Features

A feature is a section of the Epic's work that offers value and advances the Epic's completion. A feature should:

- Offer commercial value
- It should be estimable, meaning it should have enough detail for the team to estimate the amount of labor required to implement it
- Be compact enough to fit in one to three sprints; if it is too large, it should be divided into smaller pieces
- Be testable; you should be aware of the tests a feature must pass in order to satisfy the client

Some agile methodologies state features in a specific format. Two of these are provided below.

- **As a <user role>, I want to <desired functionality> so that I can <goal>.**

Example: As a coffee machine owner, I want to provide the recipes for three coffee types so I can offer a variety of coffee to my customer.

- **<action> <results> <object>**

Example: Input the recipe for a coffee type[1]

Writing a Feature - The statement of a feature must at the very least include the acceptance criteria, a brief description of the item of value, and the benefit of the feature (the points of quality or completion the feature must achieve, UAC for short).

D. User stories

A user story is the smallest unit of work in the agile framework which is also the building block of larger agile frameworks like epics and initiatives. User stories are informal, general explanations of features to be developed from the end user's or customer's perspective. They are usually written in simple language that outlines the desired outcome.

User stories are created at the start of the sprint by the developer after which the development team reviews the created story and assigns points to each story. These points indicate the complexity and effort required to complete the user story. Based on review and effort estimations the initial story might be broken down into several stories for ongoing and upcoming sprints.

User stories are often expressed in simple sentence, structured as: "**As a [persona], I [want to], [so that].**"

For example:

- As an analyst, I want to be able to login into the sales application, so that I can review sales data.
- As a customer, I want to initiate a money transfer from the mobile application, so that I don't have to visit the bank to transfer money.

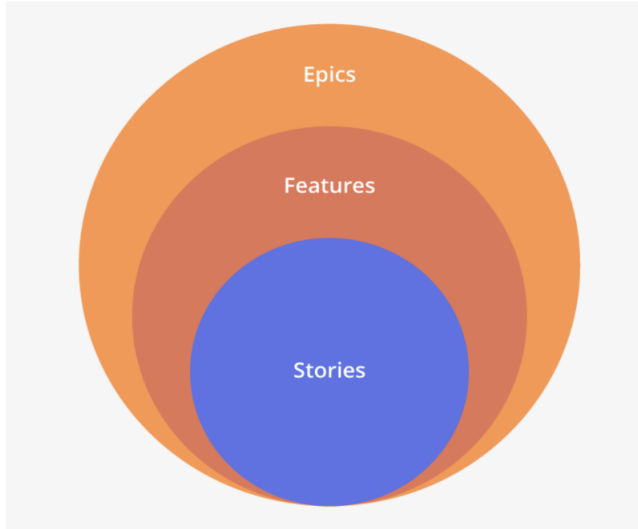
While creating user stories, one should consider following below:

- **Size** - Size of user story should be such that it can be completed within single sprint which usually last for 2 weeks.
- **Outline sub-task or tasks** - Breaking down user story into sub-task helps development team understand the complexity and efforts required to complete the user story.
- **Define exit criteria** - Exit criteria is the set of requirement for user story to be marked as completed.[4]

Advantages of user stories:

- With each completing story, the development team enjoys a small challenge and a small win which drives momentum.
- User story keep development team focused on solving problems for real users.
- Stories encourages the team to think critically and creatively about best possible way to meet end goal.
- With each story, development team gets to know more about the feature, epic and ultimately the initiative.

IV. RELATIONSHIP BETWEEN EPICS, FEATURES AND STORIES



V. BENEFITS OF ADAPTIVE PROJECT MANAGEMENT

- **Customer satisfaction** - In an agile environment, customers are kept informed about the development progress and their opinions are valued. Since the stakeholders are involved in every iteration, the outcome is likely to be inlined with the customer's expectations. This also increases customer retention.
- **Improved quality** - Agile adopts an iterative approach to development which means processes are improved for every iteration. This leads to improved quality of the project artifacts.
- **Adaptability** - Since development is done iteratively, even the last-minute changes to requirements can also be included and planned for the upcoming iterations.[3]
- **Reduced risk** - The developer team regularly evaluates project progress which means they have better visibility of forthcoming challenges and obstacles. These can be tracked and planned beforehand reducing the risk of escalations.

VI. COMPARISON BETWEEN TRADITIONAL PROJECT MANAGEMENT AND ADAPTIVE PROJECT MANAGEMENT

Characteristics	Agile approach	Traditional approach
Organizational structure	Iterative	Linear
Scale of projects	Small and medium scale	Large-scale
User requirements	Interactive input	Predefined
Involvement of clients	High	Low
Development model	Evolutionary delivery	Life cycle
Customer involvement	Customers are involved throughout project life cycle	Customers are only involved during early stage of project
Model preference	Favors adaption	Favors anticipation
Reviews and approvals	Performed after every iteration	Excessive reviews and approvals by leaders

VII. CONCLUSION

Organizations are embracing continuous change based on the capacity to sense and respond instantly to customers changing requirements. As traditional project management is not much efficient to manage the inevitable changes that occur during several projects. On the other hand, adaptive project management is well-equipped to aid project managers and software projects in managing scope, risk, and budgets. It advocates for creating a successful and valuable product.

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